Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/01/2004
PI Name:	Urbach, Jeffrey S Ph.D.		
Project Title:	Non-equilibrium Dynamics and Statistical Mechanics of Excited Granular Media in the Absence of Gravitational Barriers		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	urbach@physics.georgetown.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	202-687-6594
Organization Name:	Georgetown University		
PI Address 1:	506 Reiss Science Building		
PI Address 2:			
PI Web Page:			
City:	Washington	**State:**	DC
Zip Code:	20057-1227	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Wilkinson, R Allen	**Contact Phone:**	216-433-2075
Contact Email:	R.A.Wilkinson@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Egolf, David (Georgetown University)		
Grant/Contract No.:	NNC04GA63G		
Performance Goal No.:			
Performance Goal Text:			
	In reduced gravity or microgravity environments, regimes without enduring contacts between particles are common. The gravitational barriers present on Earth tend to mask the effects that will dominate in a granular gas. We are undertaking a series of matched experiments and computer simulations on vibrated granular media in a geometry chosen to minimize the effects of gravity consisting of a collection of identical spherical particles confined between two parallel vibrating plates. This system is well suited to investigate many of the complex non-equilibrium effects observed in granular gases. Investigate the statistical mechanics of vibrated, unsheared monodisperse granular layers. Statistical characterization essential for developing continuum models of granular flows. We will measure the energy flow from the driving to the kinetic energy of the particles, diffusive transport of particles, and the nature of observed phase transitions.		

Task Description:	We will also construct a granular rheometer that incorporates an oscillating disk into the center of the vibrating container to measure the stress tensor for the granular fluid and measure changes in local average and fluctuating quantities induced by the imposed shear. This will allow stringent tests of the assumptions and predictions of continuum equations for granular flow, particularly because we will be able to control the shear rate and the kinetic energy of the particles independently. We will also use rheometery of dense shear flow to investigate the utility of some newly developed, potentially powerful and far reaching ideas from non-equilibrium statistical mechanics. We will investigate a newly discovered segregation mechanism in mixtures of particles of different sizes, similar to the depletion force observed in colloidal suspensions. This interaction, which is masked by gravitational effects in most terrestrial granular flows, could have significant implications for flows in microgravity. The closely matched experiments and computer simulations are designed to have a broad impact on our understanding of granular flow. By carefully investigating a well controlled model system, we will be able to develop and validate kinetic and hydrodynamic descriptions of granular media that incorporate the complex correlations observed in granular flow, characterize new mechanisms for interactions and segregation in granular mixtures, and test new fundamental concepts in non-equilibrium statistical mechanics that may have wide applicability for systems driven far from equilibrium.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We are investigating the dynamics of vibrated granular layers to understand the promises and challenges of advancing the state of the art in modeling granular flow. We have discovered a freezing transition from a homogeneous, disordered liquid to an ordered solid with square symmetry coexisting with a surrounding liquid. A possible mechanism for the transition comes from an essentially identical transition that is observed in confined hard-sphere colloidal suspensions in equilibrium. That transition is well understood, and is driven by entropy maximization. The presence of solid phases with the same symmetry in both systems (different than the symmetry one would naively expect) that occur with the same geometric constraint and at a similar density strongly suggest that they share a common mechanism. Unlike the equilibrium system, however, we find that the coexisting phases have dramatically different granular temperatures (average kinetic energies per particle). This shows that the ``zeroth law'' of thermodynamics is not followed by the granular temperature. The significance of these results is that they indicate that the driving mechanism behind an equilibrium phase transition may still operate far from equilibrium, but that models of transitions in granular flow must take into account the lack of equipartition in the kinetic energy of the particles. We have completed construction of a granular rheometer and are beginning testing. The rheometer consists of a computer-controlled high precision servo motor that drives a disk in the center of the vibrating plate. Imbedded in the outside of the disk are 171 stainless steel ball bearings. We are now measuring profiles of average velocities and granular temperatures for uniformly sheared vibrated layers. We have studied an attractive interaction between large particles in a background of fluctuating small spheres. The interaction is analogous to the depletion force observed in colloidal suspensions, and represents a new mechanism of segregation in granular flows. Careful measurements of the statistical properties of the system show that some of the subtle features of the depletion interaction are present in the granular system as well. We are using this interaction to probe the utility of extensions of equilibrium statistical mechanics to far-from-equilibrium granular flows.
Bibliography Type:	Description: (Last Updated: 12/21/2007)
Articles in Peer-reviewed Journals	Prevost, A.; Melby, P.; Egolf, D.A; Urbach, J.S; "Non-equilibrium two-phase coexistence in a confined granular layer" Physical Review E , Sep-2004
Presentation	Urbach, J.S; Prevost, A.; Melby, P.; Egolf, D.A; "Non-equilibrium two-phase coexistence in a confined granular layer" Southern Workshop on Granular Materials, Pucon, Chile Dec-2003
Presentation	Melby, P; Prevost, A; Egolf, D; Urbach, J.S; "Entropy driven dynamics in vibrated granular media" Understanding Complex Systems Symposium, University of Illinois at Urbana-Champaign. May-2004